

Revised Load Interconnection System Impact Study Report

For Queue Project Q22-2107

West Middleton 138 kV
January 17, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2107. The Interconnection Customer proposes a 907.0 MW large load (data center) facility interconnecting at the West Middleton 138 kV substation in the Pennsylvania area. The request was received on June 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 568 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$110,027,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Elroy Substation / Centerpoint 138 kV	W Middleton / Lochbuie 138 kV	413.1	181	67.3
Centerpoint / Livingston Ridge Substation 138 kV	Lochbuie / West Middleton 138 kV	352.0	202	67.3
Lochbuie / W Middleton 138 kV	Centerpoint / Elroy 138 kV line	348.9	290	92.1
Livingston Ridge / W Middleton 138 kV	Lochbuie / W Middleton 138 kV	340.9	273	100.0
Lowell Cogeneration Company Lp Substation / Grasmoe 138 kV	Lochbuie / West Middleton 138 kV	157.4	184	32.7
Lowell Cogeneration Company Lp / North Nashville 138 kV	Lochbuie / W Middleton 138 kV	154.7	196	32.7
Ironwood Wind Substation / Delta Uppco 138 kV	Lochbuie 345 kV/1 kV XFMR	147.6	310	40.9
Cosmopolis / Bumont 138 kV	W Middleton / Lochbuie Substation 138 kV	141.0	210	32.7
Lochbuie / Ironwood Wind 138 kV L	Lochbuie 345/1kV TX	137.7	323	40.9
Grasmoe Substation / Buxmont 138 kV	Lochbuie / W Middleton 138 kV Line	136.5	194	32.7
Bandera Rd Substation / Sangar 138 kV	Lochbuie / West Middleton 138 kV	127.5	272	33.3
Sangar / Elroy 138 kV	Lochbuie / W Middleton 138 kV	123.6	268	33.3
Lochbuie 138 kV/1 kV Auto	Owatonna / Moser Generating Station 345 kV	119.6	533	33.4
Delta Uppco / Delmar 138 kV L	Lochbuie / Moser Generating Sta 345 kV line	118.1	343	13.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Delmar / Moser Generating Sta 138 kV	Moser Generating Station / Lochbuie 345 kV	116.1	369	13.2
Lochbuie 345/1kV Auto	Owatonna Substation / Moser Generating Sta 345 kV line	109.4	583	33.4
Lawrence Livermoe / Elroy 138 kV	Ontelaunee Energy Center / Wigwam 345 kV line	108.5	556	6.1
Lawrence Livermore / Oswego Harbor Power 138 kV	Wigwam / Ontelaunee Energy Ctr 345 kV	107.8	594	6.1
Oswego Harbor Power / Wentzville Assembly & Contiguous 138 kV	Wigwam / Ontelaunee Energy Ctr 345 kV line	107.6	601	6.1
Lochbuie / Rec Shaw 138 kV Line	Moser Generating Sta / Curecanti 345 kV Line	107.5	365	14.1
Rec Shaw / E Weymouth 138 kV	Cuecanti / Moser Generating Station 345 kV	107.2	376	14.1
Limerick / E Weymouth 138 kV	Curecanti / Moser Generating Station 345 kV	106.5	401	14.1
Wheelabrator Frackville Energy 500 kV/1 kV TX	Talenenergy Martins Creek Llc Harwood / Kline Township Cogen Facility 345 kV line	104.6	1,210	6.0
Cosmopolis / Charlotte 138 kV	Lochbuie Substation / West Middleton 138 kV	104.3	245	32.7
Wheelabrator Frackville Energy 345/1kV Transformer Bank	Talenenergy Martins Crk Llc Harwood / Kline Township Cogen Facility 345 kV	103.8	1,220	6.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Elroy / Centerpoint 138 kV	Rebuild line with higher-capacity conductor	\$3,275,000	97% (\$3,166,000)
Centerpoint / Livingston Ridge Substation 138 kV	Rebuild line with higher-capacity conductor	\$16,020,000	97% (\$15,485,000)
Lochbuie / W Middleton 138 kV	Rebuild line with higher-capacity conductor	\$1,206,000	97% (\$1,169,000)
Livingston Ridge / West Middleton 138 kV	Rebuild line with higher-capacity conductor	\$3,880,000	100% (\$3,880,000)
Lowell Cogeneration Company Lp / Grasmoe 138 kV	Rebuild line with higher-capacity conductor	\$13,490,000	93% (\$12,564,000)
N Nashville / Lowell Cogeneration Company Lp 138 kV	Rebuild line with higher-capacity conductor	\$6,937,000	93% (\$6,460,000)
Delta Uppco / Ironwood Wind 138 kV	Rebuild line with higher-capacity conductor	\$4,879,000	87% (\$4,251,000)
Cosmopolis / Bumont 138 kV	Rebuild line with higher-capacity conductor	\$7,396,000	93% (\$6,904,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lochbuie / Ironwood Wind 138 kV L	Rebuild line with higher-capacity conductor	\$2,085,000	87% (\$1,817,000)
Grasmoen Substation / Buxmont 138 kV	Rebuild line with higher-capacity conductor	\$1,568,000	93% (\$1,460,000)
Sangar / Bandera Rd 138 kV L	Rebuild line with higher-capacity conductor	\$2,635,000	97% (\$2,546,000)
Sangar / Elroy 138 kV	Rebuild line with higher-capacity conductor	\$4,019,000	97% (\$3,883,000)
Lochbuie 138/1kV Transformer	Replace transformer with higher-rated unit	\$3,638,000	89% (\$3,251,000)
Delta Uppco / Delmar 138 kV L	Rebuild line with higher-capacity conductor	\$5,534,000	53% (\$2,927,000)
Delmar / Moser Generating Sta 138 kV	Rebuild line with higher-capacity conductor	\$810,000	53% (\$428,000)
Lochbuie 345/1kV Auto	Replace transformer with higher-rated unit	\$14,169,000	89% (\$12,660,000)
Lawrence Livermoe / Elroy 138 kV	Reconductor line	\$1,427,000	95% (\$1,352,000)
Oswego Harbor Power / Lawrence Livermore 138 kV line	Reconductor line	\$2,495,000	95% (\$2,363,000)
Oswego Harbor Power / Wentzville Assembly & Contiguous 138 kV	Reconductor line	\$5,898,000	95% (\$5,587,000)
Rec Shaw / Lochbuie 138 kV L	Reconductor line	\$1,476,000	88% (\$1,302,000)
Rec Shaw / E Weymouth 138 kV	Reconductor line	\$1,286,000	88% (\$1,135,000)
Limerick / E Weymouth 138 kV	Reconductor line	\$1,770,000	88% (\$1,562,000)
Wheelabrator Frackville Energy 500 kV/1 kV TX	Replace transformer with higher-rated unit	\$21,580,000	36% (\$7,685,000)
Charlotte / Cosmopolis 138 kV	Reconductor line	\$846,000	93% (\$790,000)
Wheelabrator Frackville Energy 345/1kV Transformer Bank	Replace transformer with higher-rated unit	\$15,165,000	36% (\$5,400,000)
	Total allocated to project		\$110,027,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.